Syllabus
STAT 679: Spatial Statistics for Lattice Data - Spring 2026
University of Wisconsin - Madison

Canvas URL: https://canvas.wisc.edu/courses/501752
Course Designations and Attributes: Counts toward 50% graduate coursework requirement; Advanced

Instructional Mode & Meeting Time and Location: In-Person (Science Hall Room 110)

Number of credits: 1 credit. Learning takes place in 45 hours of learning activities, which include time in lecture presentations, R demos, homework assignments, and minute papers.

Instructor: **Mahedi Hasan**, **PhD**
Email: mhasan32@wisc.edu
Co-Instructor: Jun Zhu, Professor of Statistics
Email: junzhu@wisc.edu

Lecture Sessions:

- **Time:** Thursdays 10:00 – 11:30 am CT
- **Location:** Science Hall -110 or Zoom (On request)

Office Hours:

- Friday 2:30 – 3:00 pm CT or by appointment (in-person or Via Zoom)
- Zoom: https://uwmadison.zoom.us/j/8210069878?pwd=MFdX8CTaoKCoehZ8fAoKUemR0Wn8BX.1

Course Objective: The course is directed towards graduate students who are interested in analyzing spatial data, including students from the environmental and ecological sciences, urban and regional planning, soil sciences, plant and animal sciences, and statistics. Similar to STAT 679-I Foundations and STAT 679-II Regression & Big Data, this course will focus mostly on methods, although an understanding of concepts and of the underlying ideas will be stressed (as opposed to simply providing a cookbook of statistical formulas).

Requisites: To take the course, students must have STAT 679-I Foundations and STAT 679-II Regression & Big Data; or consent of instructor.

Learning Outcomes After completing Stat 679, a student should be able to:

- Demonstrate foundational concepts in spatial statistics
  - Classify spatial data into geostatistical, areal, and point pattern data
  - Summarize spatial data by appropriate graphs and statistics
  - Recognize spatial data as realizations of a random process
- Understand foundational spatial regression concepts
  - Apply rudimentary spatial regression models
  - Construct hypothesis tests for regression coefficients
  - Evaluate model assumptions
  - Select models based on appropriate criteria
- Perform spatial regression analysis and interpret results for spatial data
  - Use a statistical software package to aid in the analysis of spatial data

- Draw conclusions based on hypothesis tests and model selection
- Communicate the results in the context of the study questions

Homework Assignments: There will weekly homework assignments. Homework assignments should be well organized and reasonably neat. It is required that you show your work in order to receive credit. ***Late homework assignments will be penalized by 10% for each late day***. If there are extenuating circumstances, prior arrangements must be made in such cases. The importance of the exercises and homework assignments cannot be overemphasized. Much of your learning will take place while working the homework problems. Homework assignments will be due on Mondays by 11:59pm CT, unless noted otherwise.

- **Minute Papers:** Minute papers are due every Thursday by 11:59pm CT, unless noted otherwise. They are a time for you to reflect on the week's materials and your own learning. Minute papers will be graded for completion and will inform topics covered.
  - Minute Papers make up 10% of your final grade.

Exam and Project: There will be no in-class exam nor project. There will instead by a cumulative final homework assignment.

- **Final Homework Assignment:** The purpose of the cumulative final homework assignment is to review the topics covered and summarize the course. Consider this final assignment as a review of what you have learned throughout the course and the opportunity to demonstrate your understanding and mastery of the material.

Grading: The course will be for 1 credit, graded on an A-F scale. The homework assignments will count 100%

Due	Assignment	Points	Percentage %
Apr- 05 (Monday)	1. Minute Paper 1	5	72%
Apr- 05 (Monday)	2. Homework Assignment #1	35	
Apr- 10 (Friday)	1. Minute Paper 2	5	
Apr- 13 (Monday)	2. Homework Assignment #2	35	
Apr- 17 (Friday)	1. Minute Paper 3	5	
Apr- 20 (Monday)	2. Homework Assignment #3	45	
Apr- 24 (Friday)	1. Minute Paper 4	5	
Apr- 27 (Monday)	2. Homework Assignment #4	45	
May- 01 (Friday)	1. Minute Paper 5	5	28%
May- 04 (Monday)	2. Final Homework Assignment # 5	65	
	TOTAL:	250	100%

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Recommended Texts:

- Cressie, N. (1993). Statistics for Spatial Data (revised edition). Wiley, New York.
- Schabenberger, O. and Gotway, C.A. (2005). Statistical Methods for Spatial Data Analysis. Chapman and Hall, Boca Raton.
- Bivand, R., Pebesma, E., Gómez-Rubio, V (2013). Applied Spatial Data Analysis with R. Springer New York.
- Chi, G. and Zhu, J. (2019). Spatial Regression Models for the Social Sciences. SAGE Publications, Inc.

Course Schedule (subject to change)**:**

Module	Topic	Receive Back	Posted on Canvas
1	Intro, Spatial Neighbors, and Join Tests		-
			Mar- 26 (1) Homework Assignment #1
2	Moran's I and Geary's c	Apr - 05 Homework Assignment #1	Apr- 09 (1) Minute Paper
			Apr- 09 (1) Homework Assignment #2
3	Exploratory Spatial Data Analysis	Apr- 13 Homework Assignment #2	Apr- 16 (1) Minute Paper
			Apr- 16 (1) Homework Assignment #3
4	Spatial Linear Models	Apr- 20 Homework Assignment #3	Apr- 23 (1) Minute Paper
			Apr-23 (2) Homework Assignment #4
5	Spatial Linear Models (cont'd)	Apr- 27 Homework Assignment #4	Apr- 30 (1) Minute Paper
			Apr- 30 (2) Final Homework Assignment #5

Policies:

Syllabus Subject to Change Notice: This syllabus is a statement of intent and serves as an implicit agreement between the instructor and the student of this course. Every effort will be made to avoid changing anything represented in it, but the possibility exists that

unforeseen events will make syllabus changes necessary. Therefore, all material, assignments, and deadlines are subject to change. It is the student's responsibility to be aware of all course timelines, announcements, and communications from their instructor pertaining to changes in course assignments and due dates.

Graduate School Guidelines: Please refer to the Graduate School's Academic Policies & Procedures website.

Academic Integrity: By enrolling in this course, each student assumes the responsibilities of an active participant in UW-Madison's community of scholars in which everyone's academic work and behavior are held to the highest academic integrity standards. Academic misconduct compromises the integrity of the university. Cheating, fabrication, plagiarism, unauthorized collaboration, and helping others commit these acts are examples of academic misconduct, which can result in disciplinary action. This includes but is not limited to failure on the assignment/course, disciplinary probation, or suspension. Substantial or repeated cases of misconduct will be forwarded to the Office of Student Conduct & Community Standards for additional review. For more information, please refer to the website on UW-Madison Academic Integrity website.

Accommodations for Students with Disabilities: The University of Wisconsin–Madison supports the right of all enrolled students to a full and equal educational opportunity. The Americans with Disabilities Act (ADA), Wisconsin State Statute (36.12), and UW-Madison policy (Faculty Document 1071) require that students with disabilities be reasonably accommodated in instruction and campus life. Reasonable accommodations for students with disabilities is a shared faculty and student responsibility. Students are expected to inform instructors of their need for instructional accommodations by the end of the third week of the semester, or as soon as possible after a disability has been incurred or recognized. The instructor will work either directly with the student or in coordination with the McBurney Center to identify and provide reasonable instructional accommodations. Disability information, including instructional accommodations as part of a student's educational record, is confidential and protected under FERPA. Questions regarding accommodations should be directed to the McBurney Disability Resource Center website

Diversity & Inclusion: For the institutional statement on diversity and more resources, please refer to the UW-Madison Diversity and Inclusion website.